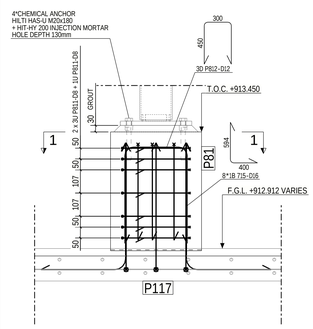
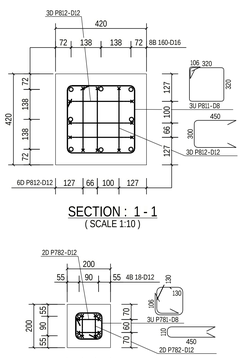
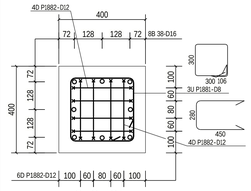


FOUNDATION P81 REINFORCEMENT
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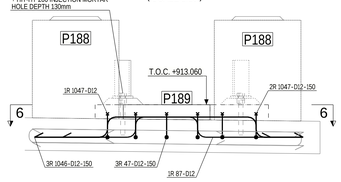


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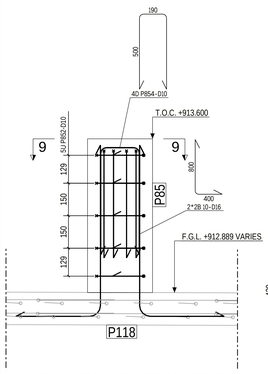
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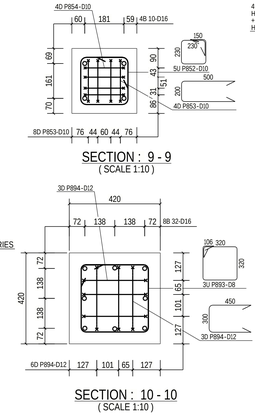
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LADDER PLATFORM P189 REINFORCEMENT
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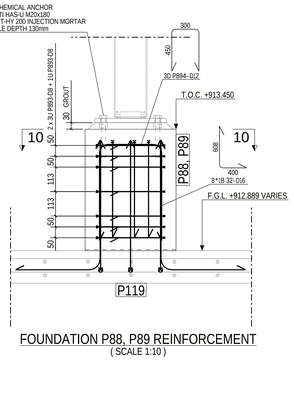


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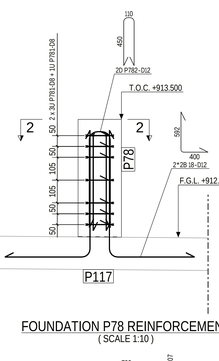


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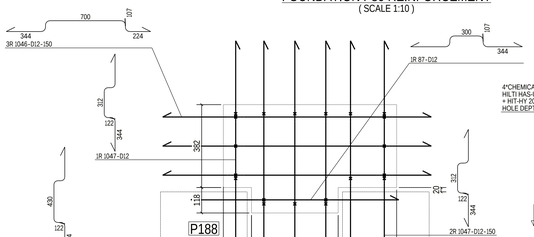
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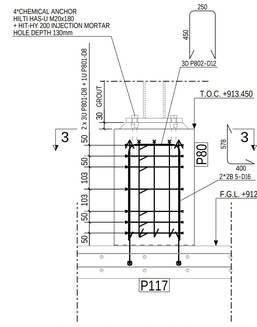
FOUNDATION P88, P89 REINFORCEMENT
(SCALE 1:10)



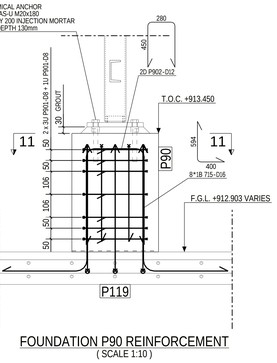
FOUNDATION P78 REINFORCEMENT
(SCALE 1:10)



(SCALE 1:10)



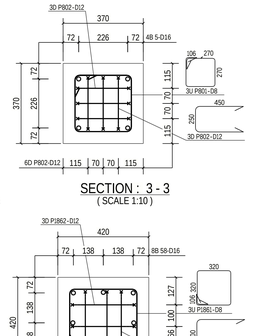
FOUNDATION P80 REINFORCEMENT
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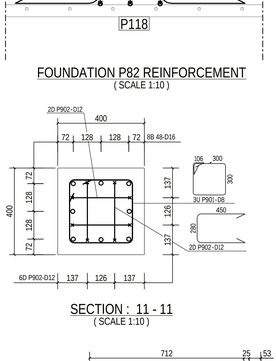
FOUNDATION P90 REINFORCEMENT
(SCALE 1:10)

(SCALE 1:10)

FOUNDATION P91 REINFORCEMENT
(SCALE 1:10)

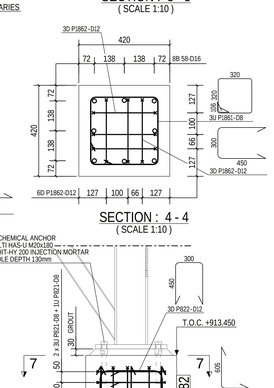


FOUNDATION P82 REINFORCEMENT
(SCALE 1:10)



(SCALE 1:10)

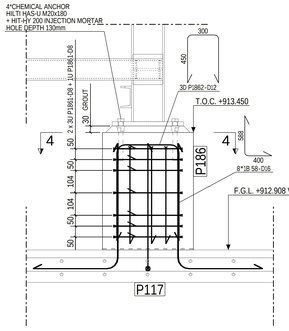
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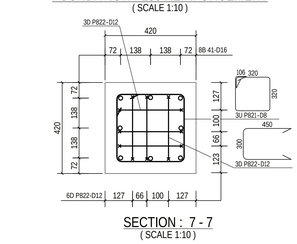
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(SCALE 1:10)

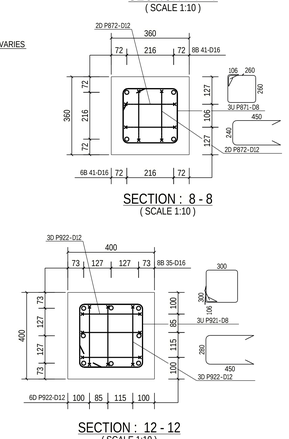
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FOUNDATION P186 REINFORCEMENT
(SCALE 1:10)

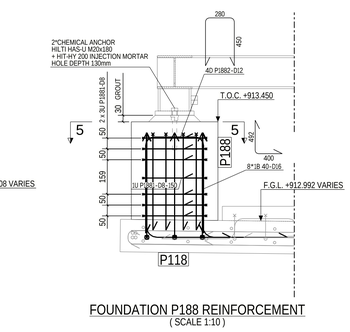


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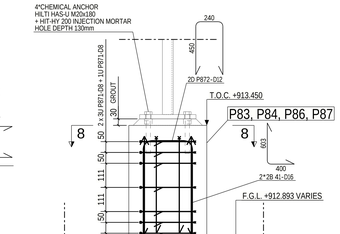


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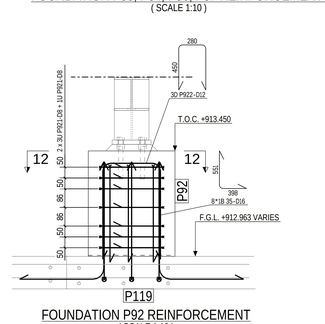
FOUNDATION P92 REINFORCEMENT
(SCALE 1:10)



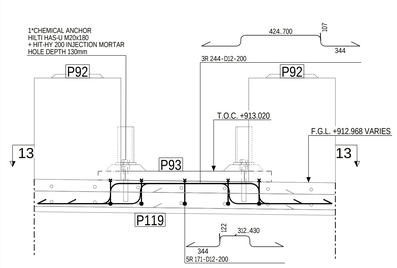
FOUNDATION P188 REINFORCEMENT
(SCALE 1:10)



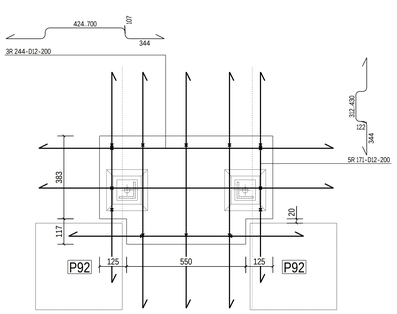
FOUNDATION P83, P84, P86, P87 REINFORCEMENT
(SCALE 1:10)



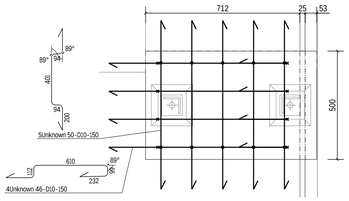
FOUNDATION P187 REINFORCEMENT
(SCALE 1:10)



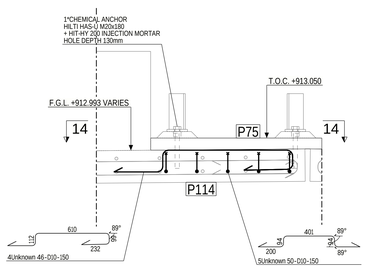
LADDER PLATFORM P93 REINFORCEMENT
(SCALE 1:10)



(SCALE 1:10)



(SCALE 1:10)



LADDER PLATFORM P75 REINFORCEMENT
(SCALE 1:10)

LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

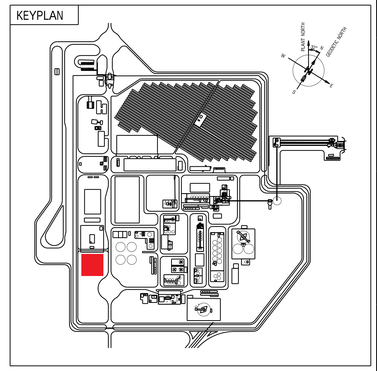
NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

NOTE: LOCATIONS OF STARTER BARS ARE SHOWN IN DRAWINGS:
MD-216-49W1-EG-CV-DED-1029
MD-216-49W1-EG-CV-DED-1030
MD-216-49W1-EG-CV-DED-1030
AND NOT SHOWN IN REINFORCEMENT LIST

LOCATIONS FOR STRUCTURES IN GENERAL ARRANGEMENT DRAWINGS:
MD-216-49W1-EG-CV-GAD-1001
MD-216-49W1-EG-CV-GAD-1002

KEYPLAN



CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	50x10mm

REINFORCING BAR LIST — CAST UNIT CODE: P75 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P751	14	B500B	8	630		3.5	106	130	130	130					32	
D	P752	8	B500B	12	950		6.7	450	110	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 10.3

REINFORCING BAR LIST — CAST UNIT CODE: P78 — PCS: 2

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P781	14	B500B	8	630		3.5	106	130	130	130					32	
D	P782	8	B500B	12	950		6.7	450	110	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 10.3

REINFORCING BAR LIST — CAST UNIT CODE: P80 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P801	7	B500B	8	1190		3.3	106	270	270	270					32	
D	P802	6	B500B	12	1090		5.8	450	250	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.1

REINFORCING BAR LIST — CAST UNIT CODE: P81 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P811	7	B500B	8	1390		3.9	106	320	320	320					32	
D	P812	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P82 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P821	7	B500B	8	1390		3.9	106	320	320	320					32	
D	P822	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P83 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P831	7	B500B	8	1110		3.1	106	250	250	250					32	
D	P832	4	B500B	12	1070		3.8	450	230	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 6.9

REINFORCING BAR LIST — CAST UNIT CODE: P84 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P841	7	B500B	8	1110		3.1	106	250	250	250					32	
D	P842	4	B500B	12	1070		3.8	450	230	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 6.9

REINFORCING BAR LIST — CAST UNIT CODE: P85 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
D	P852	15	B500B	10	1100		10.2	150	230	230	230					40	
D	P853	12	B500B	10	1150		8.5	500	200	500						40	
D	P854	12	B500B	10	1140		8.5	500	190	500						40	

REINFORCING BAR TOTAL WEIGHT [kg] 27.2

REINFORCING BAR LIST — CAST UNIT CODE: P86 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P861	7	B500B	8	1110		3.1	106	250	250	250					32	
D	P862	4	B500B	12	1070		3.8	450	230	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 6.9

REINFORCING BAR LIST — CAST UNIT CODE: P87 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P871	7	B500B	8	1150		3.2	106	260	260	260					32	
D	P872	4	B500B	12	1080		3.8	450	240	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.0

REINFORCING BAR LIST — CAST UNIT CODE: P88 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P881	7	B500B	8	1310		3.6	106	300	300	300					32	
D	P882	8	B500B	12	1120		8.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.6

REINFORCING BAR LIST — CAST UNIT CODE: P89 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P893	7	B500B	8	1390		3.9	106	320	320	320					32	
D	P894	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P90 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P901	7	B500B	8	1310		3.6	106	300	300	300					32	
D	P902	4	B500B	12	1120		4.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.6

REINFORCING BAR LIST — CAST UNIT CODE: P91 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P913	7	B500B	8	1150		3.2	106	260	260	260					32	
D	P914	4	B500B	12	1080		3.8	450	240	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.0

REINFORCING BAR LIST — CAST UNIT CODE: P92 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P921	7	B500B	8	1310		3.6	106	300	300	300					32	
D	P922	6	B500B	12	1120		6.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.6

REINFORCING BAR LIST — CAST UNIT CODE: P93 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P186 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P1861	7	B500B	8	1390		3.9	106	320	320	320					32	
D	P1862	6	B500B	12	1140		6.3	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P188 — PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
U	P1881	7	B500B	8	1310		3.6	106	300	300	300					32	
D	P1882	8	B500B	12	1120		8.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.6

REINFORCING BAR LIST — CAST UNIT CODE: P189 — PCS: 1

REINFORCEMENT BENDING SHAPES:



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1031	SLAB SECTOR GL D-G/1-4 DETAIL- AND REINFORCEMENT DRAWING
MD-331-SMDM-EG-ST-DST-0001	CONCRETE STUCTURAL STANDARDS
MD-216-49W1-EG-CV-GAD-1003	1-33/A-E GENERAL ARRANGEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1001	1-8/E-J GENERAL ARRANGEMENT DRAWING

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1029	SLAB SECTOR GL D-G/8-12 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-DED-1030	SLAB SECTOR GL D-G/4-8 DETAIL- AND REINFORCEMENT DRAWING
MD-216-1000-EG-ST-DRT-1001	DESIGN CRITERIA
MD-216-49W1-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NanWu	LuiPiz	
C02	21.03.2022	FOUNDATION P91 ADDED	JonRen	MauNot	NanWu	LuiPiz	
C01	01.03.2022	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NanWu	LuiPiz	
A01	18.02.2022	ISSUED FOR APPROVAL	JonRen	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER — REVISION 3

OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_CTB01_49039 — DOCUMENT ID: OU100746296 — REVISION 3

DRAWN	JonRen	18.02.2022
CHECKED	MauNot	18.02.2022
ENGINEER	NanWu	18.02.2022
APPROVED	LuiPiz	18.02.2022
EMPLOYER		
SCALE	1:15	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
SEC. PLATFORMS, STAIR AND PEDESTAL FOUNDATION 4
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER: MD-216-49W1-EG-CV-DED-1039 — REV: D01